Where does the infill housing go?
Addition versus division

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Puzzle introduced and resolved

Puzzle

- Density limits have greatest deadweight loss in high-priced areas
  - Across metros Hsieh and Moretti (2019)
  - Within metros Brueckner and Singh (2020)
- But recent "missing middle" upzoning uptake - sorted within metros

Resolution

- Addition of space, sure
- But a lot of infill upzoning is "division"
  - Same space, more units
  - More valuable where poorer people live by concavity
  - With mandated home sizes, rich get better locations

Some examples from Vancouver: Dunbar vs less rich Hastings-Sunrise

$4,680,000
3349 W 19th Avenue
Dunbar • Vancouver
5 bd • 5 ba • 2885 sf • 33 × 121 ft
House

Luxmore Realty

$2,388,000
3634 W 29th Avenue
Dunbar • Vancouver
3 bd • 4 ba • 1633 sf • 33 × 130 ft
Duplex

Julien Delbassee-Leflon PREC*
RE/MAX City Realty

$2,658,000
3123 E 7th Avenue
Renfrew • Vancouver
8 bd • 8 ba • 3557 sf • 33 × 122 ft
House

Tracy Niu PREC*
Nu Stream Realty Inc.

$1,599,800
2558 William Street
Renfrew • Vancouver
3 bd • 3 ba • 1545 sf
Duplex

Sutton Centre Realty

Literature

- Ma: high-income households less harmed by large home mandates
- Studies showing negative sorting of infill takeup:
  - Vancouver laneway (Davidoff, Pavlov, Somerville, 2022)
  - ADUs in California (Marantz-Elmendorf and Kim, 2023; Brueckner and Thomaz, 2024)
  - Duplexes and triplexes in Seattle (Huang et al, 2023)
  - Portland's Residential Infill Project (RIP) (Dong and Hansz, 2019)

2x2x2 model of housing choice and sorting

- Types of households: high/low income
- Types of locations θ: East (worse), West better
- Types of homes s: duplex (worse), house better
- $U(y, \theta, s) = \underbrace{\text{concave}} u(y - p(\theta, s)) + A_\theta(\theta) + A_s(s)$
- Open city, outside option $U_(0)(y) = u(y - x) + A_0$
- Specify $u(c) = log(c)$ for starters

Some comparative statics

- Set zero amenity to East and Duplex
- By open city
  $U(\text{East,Duplex}) = \log(y - p(E, D)) = \log(y - x) + A_0$
- $p(E, D) = y - (y - x)e^{A_0}$
- $p(E, H) = y - (y - x)e^{A_0 - A_s}$
- $p(W, D) = y - (y - x)e^{A_0 - A_\theta}$
- $p(W, H) = y - (y - x)e^{A_0 - A_\theta - A_s}$
- Low types outbidding for East, need $A_\theta > A_s$, A_0, x large
- Duplex split more profitable East requires:
  - $2p(E, D) - p(E, H) - c > 2p(W, D) - p(W, H) - c$
  - c a construction cost

Contrived example

	Low Type	High Type
Income	100	200
Outside x	95	95
Outside A	2	2
Outside U	3.61	6.65
Awest	2	2
Ahouse	1.5	1.5
pED	63.05	-575.85
pWD	95	95
pEH	91.76	26.88
pWH	98.88	176.57
Revenue Delta	34.35	13.43